

## FINAL EXPLANATION: BIOLOGY GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

### FINAL EXPLANATION: BIOLOGY GRADE 10: CHEMICALS OF LIFE (FINAL EXPLANATION)

Teacher Key: Exemplar Answers (do not give to students)

#### INSTRUCTIONS GIVEN TO STUDENTS

You have finished all 6 lessons on the Chemicals of Life. Now write your own COMPLETE EXPLANATION of this driving question: How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair? Think about Eliud Kipchoge, who trains near Iten in the Rift Valley and ran 42.195 km in 7,269 s (2:01:09) at Berlin 2022. Use your class model (Lesson 1 and final), your Driving Question Board, your food test results, the amylase data and the meal comparison. Write your answer in the space under each prompt. Use numbers from the tables whenever you make a claim. Do not call a meal 'best' without citing numbers. Remember safety: goggles and a hot water bath for Benedict's and HCl tests, and never taste lab samples. Use the words from the unit, such as reducing sugar, amino acids, fatty acids and glycerol, optimum and denatured. Write in full sentences in your own words.

#### Part 1: The plain plate and the marathon

**Look at Table 1 and the Kipchoge figures. Kipchoge has a body mass of about 52 kg and uses about 1 kcal per kg per km. His marathon (42.195 km) takes 7,269 s. (a) Calculate his average pace in s per km and the energy cost of the marathon. (b) His stored glycogen is about 500 g. Using 4 kcal per g, calculate the energy it holds and compare it with the marathon cost. (c) Name the three main groups of chemicals of life taught in this unit, and give the one-line role of each in the body. (d) Explain why a plate of ugali, beans, sukuma wiki and milk is 'not plain'.**

(a)  $\text{Pace} = 7,269 / 42.195 = 172.3 \text{ s per km}$  (about 2 min 52 s per km).  $\text{Energy cost} = 52 \text{ kg} \times 1 \text{ kcal per kg per km} \times 42.195 \text{ km} = \text{about } 2,194$ , so about 2,200 kcal. (b)  $\text{Glycogen energy} = 500 \text{ g} \times 4 \text{ kcal per g} = 2,000 \text{ kcal}$ . This is slightly less than the marathon cost of about 2,200 kcal, so the stored glycogen supplies almost all of the energy needed, and glycogen is a carbohydrate store. (c) The three main groups are carbohydrates (supply and store energy), proteins (build and repair the body, and make enzymes) and lipids (store a lot of energy and insulate the body). (d) The plate is not plain because it holds all three chemicals of life. Ugali is mostly starch (26 g carbohydrate per 100 g), beans and milk give protein (beans 8 g, milk 3.3 g per 100 g), and milk, sukuma wiki cooked with oil and beans supply some lipids. These chemicals give the energy, repair materials and stored energy an athlete from Iten needs.

#### Part 2: Food tests as evidence

Use your test results from Lessons 2, 3 and 4. (a) Complete the table below by writing the reagent or test used and the colour or sign that shows a positive result for each chemical.

| Chemical       | Test or reagent | Positive result |
|----------------|-----------------|-----------------|
| Starch         |                 |                 |
| Reducing sugar |                 |                 |
| Protein        |                 |                 |
| Lipid          |                 |                 |

(b) Sugarcane juice tastes very sweet but Benedict's solution stayed blue. Explain this result and describe what you did to get a positive result. (c) Ugali stayed blue with Biuret reagent, but beans turned violet. Explain what this tells you and what it does NOT tell you about ugali. (d) Give one food that tested positive for lipids and one test result that shows maize flour has no lipid.

(a) | Chemical | Test or reagent | Positive result |

| Starch         | Iodine solution                                 | Blue-black                                                                         |
|----------------|-------------------------------------------------|------------------------------------------------------------------------------------|
| Reducing sugar | Benedict's solution, heated in a hot water bath | Blue changes to green, then orange to brick-red                                    |
| Protein        | Biuret reagent                                  | Blue changes to violet                                                             |
| Lipid          | Brown paper test (and emulsion test)            | Translucent spot that stays after drying (cloudy white layer in the emulsion test) |

(b) Sugarcane juice contains sucrose, a disaccharide that is a non-reducing sugar, so Benedict's stays blue. When we boiled it with dilute HCl, the sucrose was split into the reducing sugars glucose and fructose, and Benedict's then gave brick-red. (c) Violet with beans, milk, egg white and omena shows that protein is present in them. Ugali stayed blue because it has too little protein to detect (2.5 g per 100 g in Table 1); it does not mean ugali has no protein at all. Its main chemical is starch. (d) Groundnuts left a translucent spot that stayed after drying (and gave a cloudy white layer in the emulsion test). Maize flour left no translucent spot, so it has no detectable lipid.

### Part 3: Enzymes and digestion

Use the amylase data in Table 3 (1% starch solution plus saliva amylase; the time for iodine to

(a) The rate rises from 0 at 0 C to 0.2 at 20 C and reaches its highest value of 1.0 at 37 C. It then falls to 0 at 60 C and 80 C. The optimum temperature is 37 C, the temperature with the fastest rate. (b)  $1.0 / 0.2 = 5$ , so the reaction is 5 times faster at 37 C than at 20 C. (c) At 60 C and 80 C the enzyme is denatured: its shape is permanently changed by the heat, so it can no longer work on

stop turning blue-black).

| Temperature | Time to clear starch | Relative rate (60 / time) |
|-------------|----------------------|---------------------------|
| 0 C         | not cleared in 600 s | 0                         |
| 20 C        | 300 s                | 0.2                       |
| 37 C        | 60 s                 | 1.0                       |
| 60 C        | not cleared in 600 s | 0                         |
| 80 C        | not cleared in 600 s | 0                         |

(a) Describe the pattern in the data and name the optimum temperature. (b) Calculate how many times faster the reaction is at 37 C than at 20 C. (c) Explain why no starch was cleared at 60 C and 80 C, using the word 'denatured'. (d) At pH 6.8 the rate was 1.0 but at pH 2 and pH 12 it was 0. What does this show about pH? (e) What is an enzyme, what does amylase break starch into, and why does chewed ugali taste sweet?

starch and the iodine stayed blue-black for the whole 600 s. At 0 C the enzyme is not denatured but works too slowly to clear the starch in 600 s. (d) Enzymes work best near neutral pH (optimum pH 6.8). Extreme pH such as 2 and 12 denatures amylase, so the rate was 0. (e) An enzyme is a protein catalyst made of amino acids that speeds up a reaction, and each enzyme is specific to its substrate. Amylase breaks starch into maltose, a reducing sugar. Chewed ugali tastes sweet because saliva amylase turns the starch in ugali into maltose at body temperature of about 37 C.

#### Part 4: Which meal best supports the athlete?

Mama Wanjiru, a canteen cook in Eldoret, must choose a meal

(a) Meal C energy =  $4 \times 90 + 4 \times 39 + 9 \times 45 = 360 + 156 + 405 = 921$  kcal. Fat share =  $405 / 921 = 44\%$ . (b) Meal A has the most carbohydrate (126.25 g, compared with 90 g in C and 77 g in B). Meal C has the highest fat share (44%, compared with 18% in A

for young runners from Iten. Use the Atwater factors (Energy = 4 x C + 4 x P + 9 x F) and the Table 4 data below.

| Meal | Food                                              | Carbohydrate (g) | Protein (g) | Fat (g) | Energy (kcal) |
|------|---------------------------------------------------|------------------|-------------|---------|---------------|
| A    | Ugali 300g + beans 150g + sukuma wiki 100g + milk | 126.25           | 30.25       | 15.3    | 763.7         |

and 13% in B). (c) Claim: Meal A best supports an endurance athlete. Evidence: Meal A gives the most carbohydrate (126.25 g), adequate protein (30.25 g) and moderate fat (18% of its 763.7 kcal), while Meal C gives only 90 g carbohydrate and 44% of its energy from fat, and Meal B gives only 77 g carbohydrate and 477 kcal. Reasoning: carbohydrate (starch) is broken down by enzymes to glucose, which supplies energy and builds the glycogen store (about 2,000 kcal) that fuels most of the marathon cost of about 2,200 kcal. Protein gives amino acids for repair of muscle, and a moderate amount of fat adds stored energy. So the ugali, beans, sukuma wiki and milk meal gives the right mix of chemicals of life. No meal alone supplies the whole 2,200 kcal, so glycogen stores matter. (d) Meal C has the most energy (921 kcal) but a lot of it comes from fat (44%) and it has less carbohydrate than Meal A. The main fuel for the marathon is carbohydrate stored as glycogen, so the type of chemical matters as well as the total kcal.

|   |                                                                                                               |    |              |        |         |
|---|---------------------------------------------------------------------------------------------------------------|----|--------------|--------|---------|
|   | 2<br>5<br>0<br>g                                                                                              |    |              |        |         |
| B | Gi<br>th<br>er<br>i<br>4<br>0<br>0<br>g<br>+<br>su<br>ku<br>m<br>a<br>wi<br>ki<br>1<br>0<br>0<br>g            | 77 | 2<br>6.<br>5 | 7      | 47<br>7 |
| C | M<br>a<br>n<br>d<br>az<br>i<br>2<br>0<br>0<br>g<br>+<br>ny<br>a<br>m<br>a<br>ch<br>o<br>m<br>a<br>1<br>0<br>0 | 90 | 3<br>9       | 4<br>5 | 92<br>1 |

|  |   |  |  |  |  |
|--|---|--|--|--|--|
|  | g |  |  |  |  |
|--|---|--|--|--|--|

(a) Show the energy calculation for Meal C and the percentage of its energy that comes from fat. (b) Which meal has the most carbohydrate, and which has the highest percentage of energy from fat? (c) Write a Claim-Evidence-Reasoning answer to the question: which meal best supports an endurance athlete? Use at least three numbers from the table, and compare the meal with the marathon energy cost of about 2,200 kcal. (d) Explain why a runner should not choose a meal only because it has the most energy.

## Part 5: Putting it all together

**Write a complete final explanation that answers the driving question: How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair? In your answer: (a) name the chemical of life in each food and its role (energy, growth and repair, stored energy and insulation); (b) refer to at least two test results as evidence; (c) explain the part played by enzymes such as amylase and what affects them; (d) use at least one number from Table 1, Table 3 or Table 4; (e) say how your**

A plate of ugali, beans, sukuma wiki and milk is not plain, because it holds the chemicals of life that an athlete from Iken like Eliud Kipchoge needs. Ugali is mostly maize starch (26 g carbohydrate per 100 g). Starch is a polysaccharide, a long chain of glucose units, and it supplies and stores energy as glucose and glycogen. Iodine turned maize flour blue-black, which is evidence for starch. Beans, milk and omena contain protein, which turned Biuret reagent violet; proteins are chains of amino acids that the body uses to grow and repair muscle damaged by running, and to make enzymes. Ugali stayed blue with Biuret because it has too little protein (2.5 g per 100 g) to detect. Lipids from milk, groundnuts and cooking oil store a lot of energy (9 kcal per g, more than twice the 4 kcal per g of carbohydrate) and insulate the body; groundnuts left a translucent spot on brown paper and have 49 g fat per 100 g. These big molecules must be broken into small ones the body can absorb. Enzymes do this. Saliva amylase, a protein catalyst, breaks starch into maltose, which is why chewed ugali tastes sweet. Amylase works fastest at its optimum of 37 C and near neutral pH 6.8. At 37 C the reaction was 5 times faster than at 20 C (1.0 / 0.2), and at 60 C or at pH 2 and 12 the enzyme is denatured and the starch was not cleared in 600 s. Body temperature of about 37 C keeps the enzymes working. Kipchoge's marathon costs about 2,200 kcal and his glycogen store is about 2,000 kcal, so the carbohydrate in his food is the main fuel. Meal A (ugali, beans, sukuma wiki and milk) gives 126.25 g carbohydrate, 30.25 g protein and 18% of its 763.7 kcal from fat, so it best supports an endurance athlete compared with Meal B (77 g carbohydrate) and Meal C (90 g carbohydrate and 44% of energy from fat). My initial model only showed that ugali and beans contain energy chemicals; my final model adds the structure of carbohydrates, proteins and lipids, the food tests, enzymes with their optimum temperature and pH, and the energy numbers. So simple Kenyan foods can fuel a 2:01:09 marathon.

|                                                                     |  |
|---------------------------------------------------------------------|--|
| final class model is different from your initial model in Lesson 1. |  |
|---------------------------------------------------------------------|--|

| RUBRIC                                          |                                                                                                                                                                                                                                       |                                                                                                                         |                                                                                                |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Criterion                                       | Excellent (4)                                                                                                                                                                                                                         | Proficient (3)                                                                                                          | Developing (1–2)                                                                               |
| Understanding of chemicals of life              | Correctly names carbohydrates, proteins and lipids, gives their building blocks (sugars, amino acids, fatty acids and glycerol) and roles, and links each to Kenyan foods such as ugali, beans, milk and groundnuts.                  | Names the three groups and gives correct roles for most, with some Kenyan food examples but little detail on structure. | Names some groups but roles or foods are missing, confused or incorrect.                       |
| Use of evidence (tests and tables)              | Accurately uses food test colours and results (iodine, Benedict's, Biuret, brown paper and emulsion) and specific numbers from Tables 1, 3 and 4 to support every claim.                                                              | Uses some test results and numbers correctly, but a few claims have no evidence.                                        | Gives few or incorrect results or numbers; claims are mostly unsupported.                      |
| Enzyme explanation                              | Explains that enzymes are protein catalysts, that amylase breaks starch to maltose, and explains the optimum temperature and pH, denaturation and specificity using the rate data (for example, 5 times faster at 37 C than at 20 C). | Explains the role of amylase and the optimum conditions, with limited use of data or of the term denatured.             | States that enzymes help digestion but gives little or incorrect detail on conditions or data. |
| Explaining the phenomenon                       | Clearly explains how ugali, beans, sukuma wiki and milk supply energy, repair and stored energy for a marathon runner such as Kipchoge, linking to the 2,200 kcal cost and 2,000 kcal glycogen store.                                 | Explains most of how the plate supports performance, but the link to the marathon figures is partial.                   | Gives a general statement that food gives energy without linking chemicals to the phenomenon.  |
| Scientific reasoning (Claim-Evidence-Reasoning) | Makes a clear claim about the best meal, supports it with at least three correct Table 4 numbers, and reasons logically why carbohydrate, protein and fat shares matter.                                                              | Makes a claim with some correct numbers and partial reasoning.                                                          | Makes a claim with no numbers or reasoning, or the numbers are wrong.                          |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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